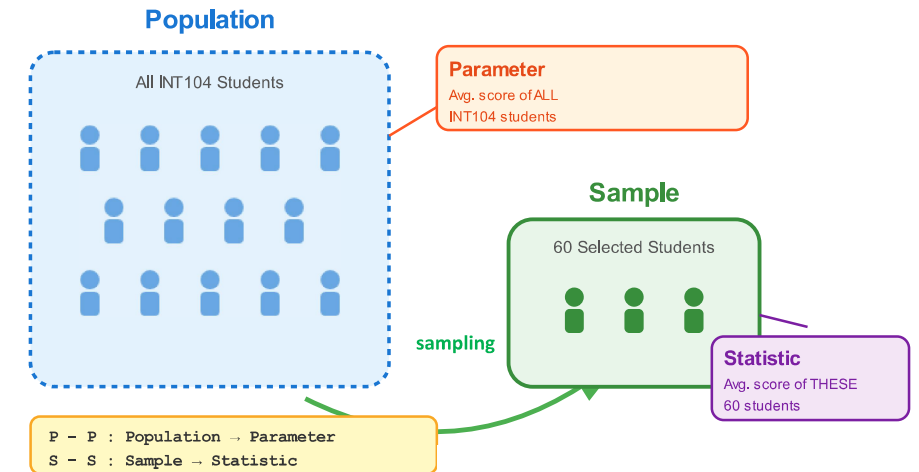


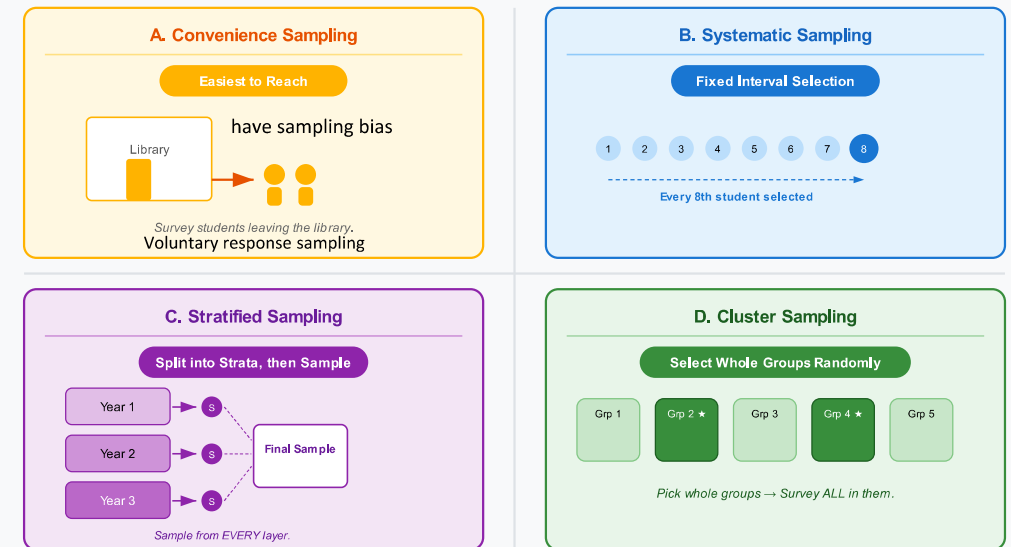
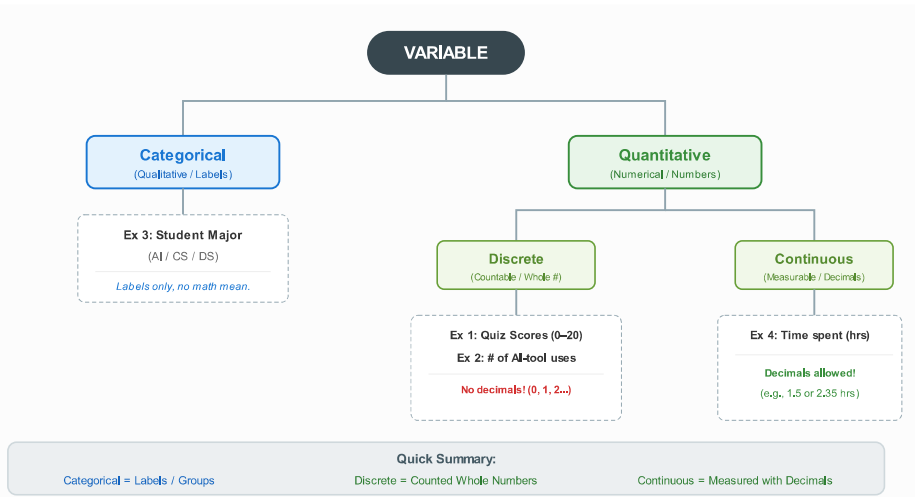
Review

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1 Basic Statistics & Variables

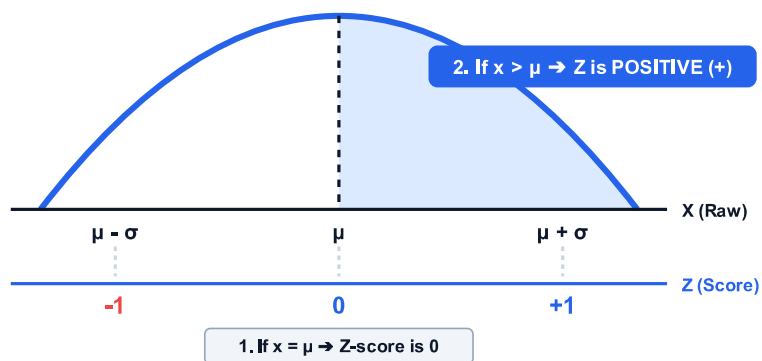


Sample method



Q8. Z-SCORE: $z = (x - \mu) / \sigma$

Mean Standard deviation



BLANK 1: THE NAME

The normal distribution is also called the **GAUSSIAN** dist.

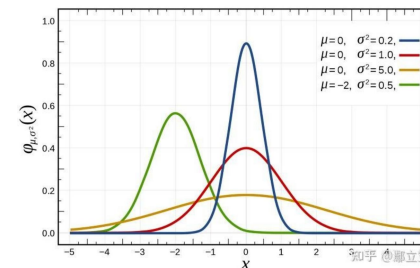
BLANK 2: THE SHAPE

The curve is **BELL**-shaped and symmetric.

BLANK 3: PARAMETERS

μ denotes the **MEAN**,

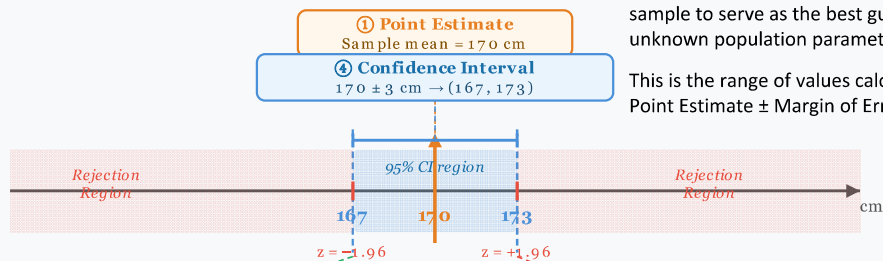
σ denotes the **STANDARD DEVIATION**



Confidence Interval — Core Concepts

This is the single value calculated from a sample to serve as the best guess for the unknown population parameter.

This is the range of values calculated as: Point Estimate $\pm$ Margin of Error



② Confidence Level
95% of intervals contain the true population mean

③ Critical Value
 $z = \pm 1.96$ for 95% CI
Defines the CI boundaries

The critical value (z) defines the edges of the "Acceptance" (Fail to Reject) zone vs. the "Rejection" zone.

This represents the probability that the interval will contain the true population mean if we were to repeat the sampling process many times.

Point Estimate CI Range Critical / Rejection Zone Confidence Level

Q3 — Confidence Level, Tails & Cumulative Area

Visualizing α , $\alpha/2$, and the cumulative area to the left of the positive critical z-value

Step-by-Step Calculation

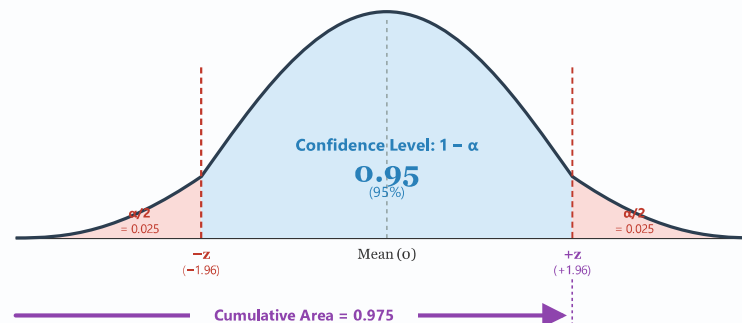
1. α (Total Error Rate) = $1 - 0.95 = 0.05$ (Both tails combined)
2. $\alpha / 2$ (Single Tail) = $0.05 \div 2 = 0.025$ (Split equally)
3. Cumulative Area = $1 - 0.025 = 0.975$

Why Cumulative Area is 0.975?

Standard Z-tables show the cumulative area to the LEFT. To find z for the right boundary, we must add up:

Left Tail (0.025) + Center (0.95) = 0.975

→ Looking up 0.975 in the Z-table yields $z = 1.96$

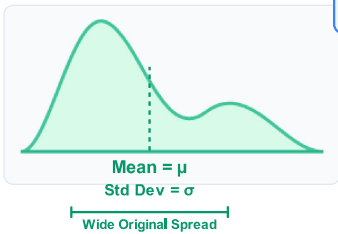


Central Limit Theorem

1. ORIGINAL POPULATION

Shape: Literally ANYTHING!

(Skewed, bimodal, messy... it doesn't matter)



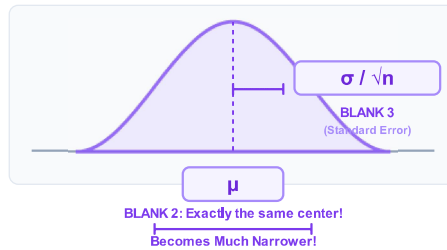
BLANK 1
 $n > 30$

Take samples of
Calculate $\bar{x}$

2. SAMPLING DISTRIBUTION of $\bar{x}$

Shape: ALWAYS A PERFECT BELL CURVE!

(This is the "Magic" of the theorem)



DETAILED EXPLANATION SUMMARY:

- When $n > 30$, the distribution of sample means ($\bar{x}$) always follows a normal distribution, regardless of the original population's shape.
- The mean of these sample means is identical to the original population mean (μ).
- The standard deviation shrinks rapidly, becoming the "Standard Error", calculated as ($\sigma / \sqrt{n}$).

Real-World Parameters

A Mean (μ)
Example:
Test the average height

B Variance (σ^2)
Example:
Test sensor error fluctuation

C Proportion (p)
Example:
Test election polling rates

Statistical Distributions

1 Normal (Z)
Symmetric standard bell curve.

2 Chi-Square (χ^2)
Right-skewed, strictly positive.

3 Student t (t)
Symmetric but with thicker tails.

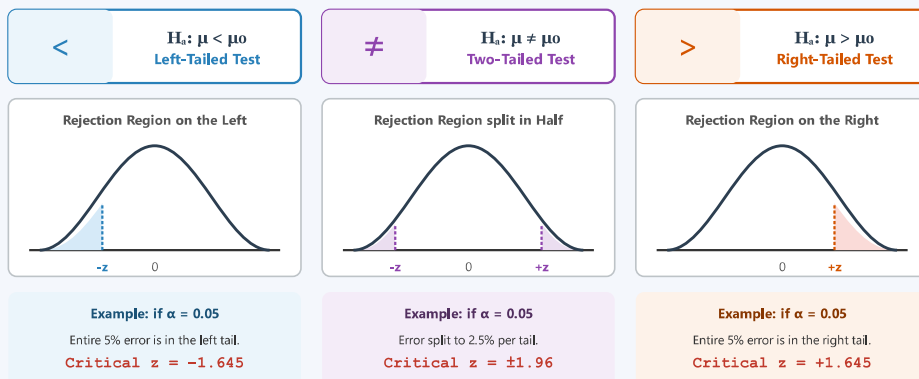
Assuming σ is unknown

Requires Large Sample

Final Answer Map: A → 3, B → 2, C → 1

Q10 — Test Type, Direction, and Critical Region

The symbol in the Alternative Hypothesis (H_a) acts as an "arrow" pointing to the rejection region.



Question Answer (Part 4) & Ultimate Takeaway:

Because the rejection region shifts and splits, *Different test types* **CRITICAL Z SCORES**

Apply the Rule: The 4 Tricky Cases

- $P = 0.031 \leq \alpha = 0.05 \rightarrow \text{Reject } H_0$
- $P = 0.120 > \alpha = 0.05 \rightarrow \text{Fail to reject}$
- $P = 0.010 \leq \alpha = 0.01 \rightarrow \text{Reject } H_0$
* Rule is " $\leq$ ", so an EXACT match rejects H_0 .
- $P = 0.0101 > \alpha = 0.01 \rightarrow \text{Fail to reject}$
* 0.0101 is slightly larger than 0.0100.

- Case 1:** $P=0.031$, $\alpha=0.05$. 0.031 is strictly less than 0.05. Because P is low, we **Reject H_0** .
 - Case 2:** 0.120 is clearly larger than 0.050. We do not have enough evidence, so we **Fail to reject H_0** .
 - Case 3:** Look at the rule carefully—it says 'less than OR EQUAL TO'. Since it's an exact tie, the rule is met. We **Reject H_0** .
 - Case 4:** 0.0101 is just a tiny fraction *larger* than 0.01. It crossed the line! Therefore, we **Fail to reject H_0** .
- Finally, for the conclusion wording: When we fail to reject the null hypothesis, we never say we 'accept' it. The legally correct statistical phrase is: 'There is **not sufficient** evidence to support/warrant rejection of the claim.'

Bayes' Rule & Naive Bayes

$$P(c|x) = \frac{P(x|c) \times P(c)}{P(x)}$$

Posterior
 $P(c|x)$

Probability of class 'c' given data 'x'.
"Updated Belief"

Likelihood
 $P(x|c)$

Probability of data 'x' if class is 'c'.
"Pattern matched"

Prior
 $P(c)$

Initial probability of class 'c' overall.
"Initial Assumption"

Evidence
 $P(x)$

Overall probability of observing data 'x'.
"Normalization Factor"

Real-World Example: Spam Email Filter

Is the email Spam (c) given it contains the word "FREE" (x)?

- 1. **Prior P(c)** : Usually, 20% of all emails you receive are spam.
- 2. **Likelihood P(x|c)** : 80% of confirmed spam emails contain the word "FREE".
- 3. **Evidence P(x)** : Regardless of type, 25% of ALL emails contain the word "FREE".
- 4. **Posterior P(c|x)** : If you see an email with "FREE", what is the chance it is spam?

Calculation: $P(c|x) = (0.80 \times 0.20) / 0.25 = 0.64$ — (64% chance it is spam!)

Step 1: Write the Likelihood Function (H=4, T=6)

$$p(\text{Data} | \theta) = L(\theta) = \theta^4 \times (1 - \theta)^6$$
 (Where θ is the unknown probability of getting Heads)

The estimated parameter (θ hat) is simply the empirical frequency:

$$\text{MLE } \hat{\theta} = k / n = 4 / 10 = 0.4$$

Likelihood Computation & Model Selection

Which coin model best explains the observed sequence?

Step 1: Observed Data (D)

T H H T T T H T T H

Count: H = 4 , T = 6

Step 2 & 3: Compute Likelihood $L = P(\text{Data} | \text{Model})$

Model 1 (Fair Coin)

$P(H) = 0.5$, $P(T) = 0.5$

$L_1 = (0.5)^4 \times (0.5)^6$

$L_1 = (0.5)^{10} = 1 / 1024$

$L_1 \approx 0.000976$

vs

Model 2 (Biased Coin)

$P(H) = 0.4$, $P(T) = 0.6$

$L_2 = (0.4)^4 \times (0.6)^6$

$L_2 = 0.0256 \times 0.046656$

$L_2 \approx 0.001194$

Conclusion: Model 2 is Selected

Because $L_2 (0.001194) > L_1 (0.000976)$. The data is more likely under Model 2.

MAP vs. MLE: The Tale of Two Estimators

How do we choose the best hypothesis 'c' given our data 'x'?

MLE

(Maximum Likelihood Estimation)

Maximize: $p(x | c)$

Focus:

"Which class 'c' makes this exact data 'x' most likely to appear?"

Data-Driven Only (Ignores Prior)

MAP

(Maximum A Posteriori)

Maximize: $p(c | x)$

Focus:

"Given this data 'x', which class 'c' is most probable in reality?"

Data + Prior Experience

The Mathematical Relationship (Bayes' Theorem)

$$p(c | x) \propto p(x | c) \times p(c)$$

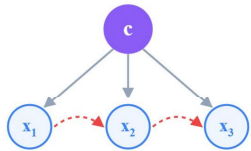
MAP (Posterior) is proportional to MLE (Likelihood) times the Prior.

The "Naive" Assumption of Naive Bayes

Features are conditionally independent given the class c.

Real World (Complex)

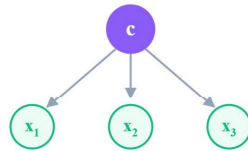
Features are often correlated



Messy Interdependencies

Naive Bayes (Simplified)

Features are fully independent



No Cross-Talk (Independent)

The Mathematical Transformation

Instead of calculating a massive, complex joint probability, we just multiply individual probabilities!

$$\begin{aligned} &P(x_1, x_2, x_3 | c) \\ &\text{becomes} \\ &P(x_1 | c) \times P(x_2 | c) \times P(x_3 | c) \end{aligned}$$

Naive Bayes Classification: The smoothing effect

Computing raw probabilities vs. applying Laplacian Correction (+1)

Target Query (x):

Outlook = Overcast, Temp = Cool, Humidity = Normal, Windy = True

Part A: No Smoothing

Score S_Yes:

$$\begin{aligned} &= P(\text{Yes}) \times P(\text{Overcast} | \text{Yes}) \times P(\text{Cool} | \text{Yes}) \dots \\ &= (9/14) \times (4/9) \times (3/9) \times (6/9) \times (3/9) \\ &\approx 0.02116 \end{aligned}$$

Score S_No:

$$\begin{aligned} &= P(\text{No}) \times P(\text{Overcast} | \text{No}) \times P(\text{Cool} | \text{No}) \dots \\ &= (5/14) \times (0/5) \times (1/5) \times (1/5) \times (3/5) \\ &= 0 \end{aligned}$$

⚠ The "Zero Veto" wiped out the probability.

Part B: Laplacian Smoothing

Score S_Yes:

$$\begin{aligned} &\text{Conditionals} = (\text{Count} + 1) / (\text{Total } N + |V|) \\ &= (9/14) \times (5/12) \times (4/12) \times (7/11) \times (4/11) \\ &\approx 0.02066 \end{aligned}$$

Score S_No:

$$\begin{aligned} &\text{Rescuing the 0 count using add-one:} \\ &= (5/14) \times (1/8) \times (2/8) \times (2/7) \times (4/7) \\ &\approx 0.00182 \end{aligned}$$

✅ The absolute zero is fixed to a realistic 1/8.

Final MAP Predictions

Part A Winner:
YES (0.02116 > 0)

Part B Winner:
YES (0.02066 > 0.00182)

The Zero-point Problem & Laplacian Smoothing

How to prevent unseen events from destroying entire probability models.

1. Unsmoothed MLE (Maximum Likelihood Estimation)

Raw Counts: N(a) = 2, N(b) = 0, N(c) = 1 Total N = 3

$$P(a) = \frac{2}{3}, \quad P(b) = \frac{0}{3} = 0, \quad P(c) = \frac{1}{3}$$

⚠ THE "ZERO VETO": In Naive Bayes, multiplying by P(b)=0 will make the entire final probability exactly 0!

2. Laplacian Smoothing (Add-One)

$$P(\text{event}) = \frac{\text{Count} + 1}{\text{Total } N + |V|} \quad |V| = \text{Number of categories}$$

If we add 1 to every count:

New N(a) = 2 + 1 = 3
New N(b) = 0 + 1 = 1
New N(c) = 1 + 1 = 2

Notice the Total N:

We have 3 categories: V = {a, b, c}, so |V| = 3
New Denominator = Total N (3) + |V| (3) = 6

$$\text{Smoothed Posteriors:} \quad P(a) = \frac{3}{6} = 1/2, \quad P(b) = \frac{1}{6}, \quad P(c) = \frac{2}{6} = 1/3$$

Correlation & Linear Regression

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2 - (\sum x)^2)} \sqrt{n(\sum y^2 - (\sum y)^2)}}$$

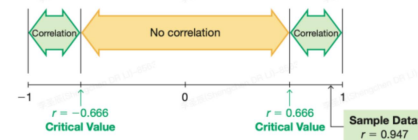


FIGURE 10-3 Critical r Values and the Computed r Value

Computing Pearson Correlation (r)

Breaking down the summary statistics into 3 manageable chunks

Data: n = 5, $\sum x = 10$, $\sum y = 20$, $\sum x^2 = 30$, $\sum y^2 = 90$, $\sum xy = 47$

1. Numerator (Top)

$$\begin{aligned} &n(\sum xy) - (\sum x)(\sum y) \\ &= 5(47) - (10)(20) \\ &= 35 \end{aligned}$$

2. Denominator (X Part)

$$\begin{aligned} &[n(\sum x^2) - (\sum x)^2] \\ &= [5(30) - 10^2] \\ &= 50 \end{aligned}$$

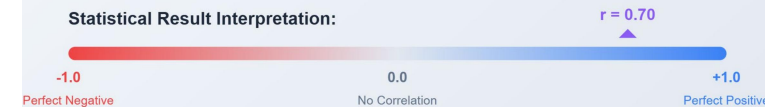
3. Denominator (Y Part)

$$\begin{aligned} &[n(\sum y^2) - (\sum y)^2] \\ &= [5(90) - 20^2] \\ &= 50 \end{aligned}$$

Pearson correlation coefficient r is primarily used to measure
The linear relationship between two variables

$$r = \frac{35}{\sqrt{(50 \times 50)}} = \frac{35}{50} = 0.70$$

Statistical Result Interpretation:



1. "r is always between -1 and +1 (inclusive)."

✓ TRUE

r is standardized (Z-scores). It cannot escape the bounds of reality.

-1.0 (Limit)

0.0 (No Linear Fit)

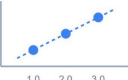
+1.0 (Limit)

2. "If all X are scaled (multiplied), r changes."

✗ FALSE

r is logically **Unitless / Dimensionless**. Scaling X does not change the core relationship.

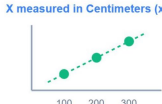
X measured in Meters



1.0 2.0 3.0

EQUALS

X measured in Centimeters (x100)



100 200 300

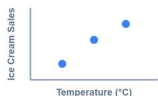
r = 0.95 is identical

3. "Interchanging x and y changes r."

✗ FALSE

r is **Symmetrical**. The relationship between X and Y is identical to the relationship between Y and X.

X-Y Relationship



Ice Cream Sales

Temperature (°C)

EQUALS

Y-X Relationship (Swapped)



Temperature (°C)

Ice Cream Sales

r = 0.85 is identical

Regression Coefficients from r (Calculation)

📊 GIVEN SUMMARY STATISTICS

Correlation (r) = 0.60 | Mean ($\bar{x}$)=10, ($\bar{y}$)=100
Std Dev (sx)=2 | Std Dev (sy)=5

🔍 CORE FORMULAS

Slope: $b_1 = r \times (s_y / s_x)$
Intercept: $b_0 = \bar{y} - (b_1 \times \bar{x})$
Equation: $\hat{y} = b_0 + b_1x$

🎯 REQUIRED TASKS

1. Calculate the slope (b_1) value
2. Calculate the intercept (b_0) value
3. Construct the final regression equation

STEP 1: FIND SLOPE (b_1)

$$b_1 = 0.60 \times (5 / 2)$$
$$b_1 = 0.60 \times 2.5$$
$$b_1 = 1.5$$

Interpretation:

For each +1 unit in X, Y increases by 1.5 units.

STEP 2: FIND INTERCEPT (b_0)

$$b_0 = 100 - (1.5 \times 10)$$
$$b_0 = 100 - 15$$
$$b_0 = 85$$

Interpretation:

Predicted value of Y when the variable X is 0.

STEP 3: BUILD EQUATION

$$\hat{y} = b_0 + b_1x$$

Substitute values:

$$\hat{y} = 85 + 1.5x$$

Summary:

This linear equation models the predictive relationship between X and Y.

FINAL RESULTS

Slope (b_1): 1.5

Intercept (b_0): 85

Regression Eq:

$$\hat{y} = 85 + 1.5x$$

Detailed Explanation: Core Properties of Pearson's Correlation Coefficient r

📖 Core Concept of r

Quantifies the strength of linear relationships between continuous variables (Range: [-1, 1]).

- $r=1$: Perfect Positive | $r=-1$: Perfect Negative
- $r=0$: No linear correlation (not necessarily no relationship).

🏆 FINAL CONCLUSION

The only correct answer is B

💡 Teaching Analogy & Tips

- Ruler Analogy: r only measures "straight lines" (linear), not curves.
- Outliers: Like bent marks on a ruler—they heavily bias the result.
- Slope vs r : Slope is scale-dependent (gradient), r is standardized.

✗ Option A: Measures any relationship (including nonlinear) [INCORRECT]

r is specifically designed for linear relationships only. It fails to capture nonlinear patterns (e.g., a perfect quadratic $y=x^2$ will yield $r \approx 0$).

✓ Option B: Measures linear strength (ignores nonlinear) [CORRECT]

This aligns with the precise definition of Pearson's r. For nonlinear relationships, alternative statistics (e.g., Spearman's rho) are required.

✗ Option C: r equals the regression line slope (b) [INCORRECT]

They share the same sign but differ in value. Slope is scale-dependent while r is a scale-invariant, unitless measure of strength.

✗ Option D: r is unaffected by outliers [INCORRECT]

r is extremely sensitive to outliers. A single extreme data point can drastically alter the value of r and even reverse the correlation direction.

Part B — One Iteration of k-means

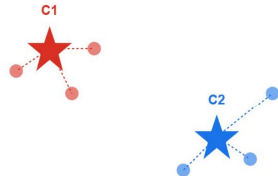
Given 7 points (from the lecture example):
(1.0, 1.0), (1.5, 2.0), (3.0, 4.0), (5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)

Let $k = 2$ and initialise centroids as:
 $c_1^{(0)} = (1.0, 1.0)$, $c_2^{(0)} = (5.0, 7.0)$.

Use Euclidean distance. If a point is equidistant, assign it to the smaller-index cluster.

1. Assign each point to Cluster 1 or Cluster 2.
2. Compute the updated centroids $c_1^{(1)}$ and $c_2^{(1)}$.

Step 1: Assign to Closest Centroid



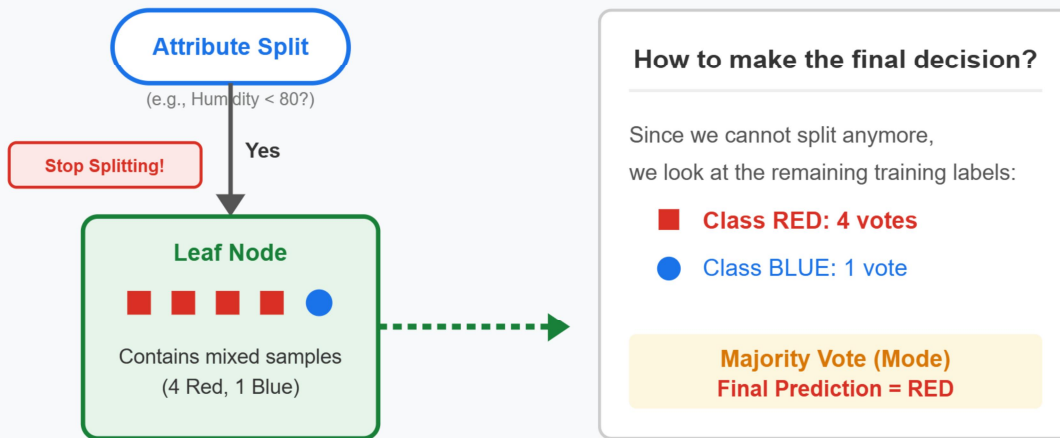
Step 2: Update Centroids (Take Average)



The star moves to the exact center of its points!

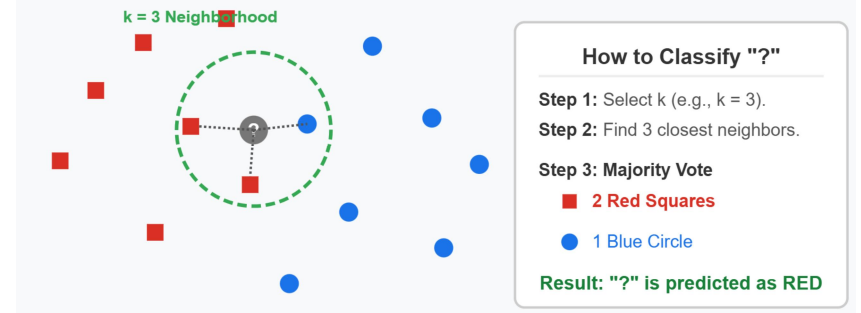
Decision Tree: How a Leaf Node Makes a Decision

When splitting stops, the leaf predicts using a Majority Vote (Mode)



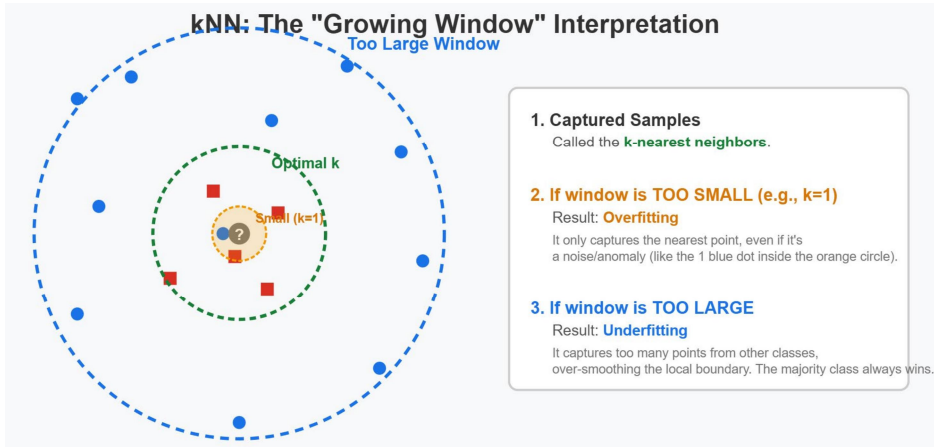
k-Nearest Neighbors (kNN) Classification Rule

The predicted label is determined by a Majority Vote



C. majority vote among the nearest neighbours' labels

kNN: The "Growing Window" Interpretation



The lecture describes a cell centred at x that grows until it captures k_n samples.

1. They are called the k_n nearest neighbours.
2. If the window is too small: **most of the volume may be empty / over-sensitive to noise.**
3. If the window is too large: **important variations may be lost.**

Margin Width Calculation

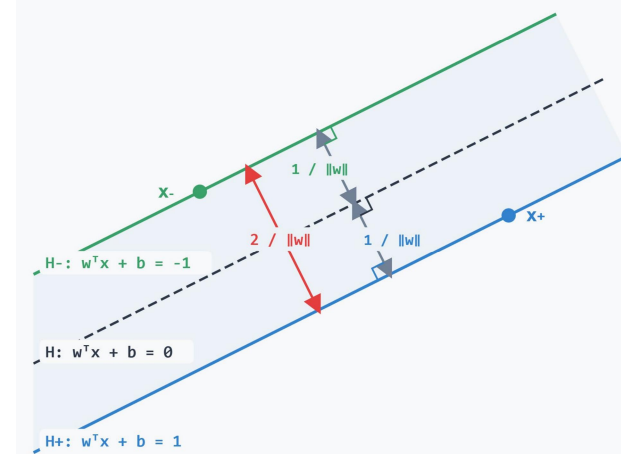
Distance from H to $H+$: $1 / \|w\|$

Distance from H to $H-$: $1 / \|w\|$

Total Street Width = $2 / \|w\|$

The Golden Rule of SVM:

To MAXIMIZE the fraction $2/\|w\|$, we must MINIMIZE the denominator $\|w\|$



Q1

$H = \{x : w^T x + b = 0\}, w \neq 0$

1. $\text{dist}\pm(x, H) = \frac{w^T x + b}{\|w\|}$

2. The sign indicates which side of the hyperplane the point lies on.

$\text{dist}\pm(x, H) > 0$

$\text{dist}\pm(x', H) < 0$

w
(Normal)

KKT Complementary Slackness (Classification of Points)

Hard-Margin Support Vector Machine (SVM)

CORE EQUATION (Complementary Slackness)

$$\alpha_i (1 - y_i (w^T x_i + b)) = 0$$

Case 1: Point is strictly OUTSIDE margin

$$y_i (w^T x_i + b) > 1$$

Term in parenthesis is not zero. To satisfy equation:

$$\alpha_i = 0$$

Case 2: Point is exactly ON margin boundary

$$y_i (w^T x_i + b) = 1$$

Constraint is active (parenthesis = 0). Therefore:

$$\alpha_i > 0$$

Definition: Support vectors are exactly the points such that $\alpha_i > 0$.

Sample Position	$y_i (w^T x_i + b)$	α_i	Support Vector?
Outside margin	> 1	$= 0$	No
On margin boundary	$= 1$	> 0	Yes

Confusion Matrix

Metric	Full Name	Definition
Recall (Sensitivity)	True Positive Rate	(1)
Fall-out	False Positive Rate	(3)
Miss rate	False Negative Rage	(2)
Specificity (Selectivity)	True Negative Rate	(4)

$$recall = \frac{TP}{TP + FN} = \frac{TP}{n^+}$$

(1)

$$fallout = \frac{FP}{TN + FP} = \frac{FP}{n^-}$$

(3)

$$miss = \frac{FN}{TP + FN} = \frac{FN}{n^+}$$

(2)

$$specificity = \frac{TN}{TN + FP} = \frac{TN}{n^-}$$

(4)

NB: Recall is also named sensitivity

Confusion Matrix

More general measurements:

$$precision = \frac{TP}{TP + FP}$$

(5)

$$accuracy = \frac{TN + TP}{TP + TN + FP + FN} = \frac{TN + TP}{n}$$

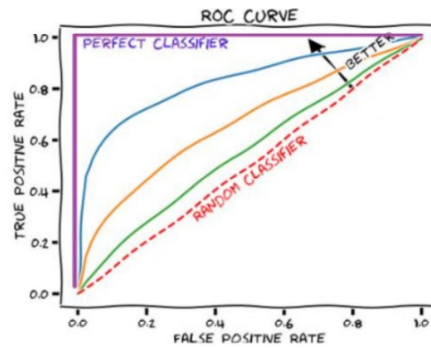
(6)

A good measurement should balance the multiple aspects such as F_1 Score (F-Measure)

$$F_1 score = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} = 2 \cdot \frac{precision \cdot recall}{precision + recall}$$

(7)

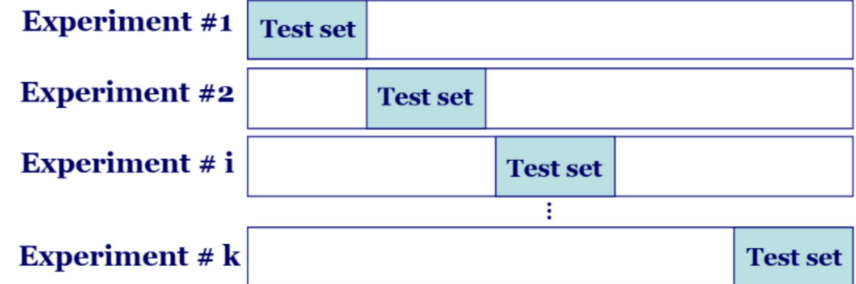
Receiver Operation Characteristics (ROC)



- One of the measurement for ROC test is Area under Curve (AUC)
- $AUC > 0.5$, the system performance is better than random guess.

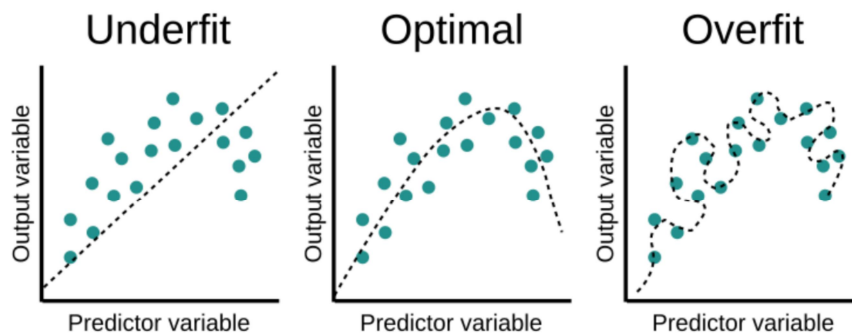
k-fold Cross Validation

- k-fold (k is usually set to 5, 10 and other numbers)
 - Randomly partition the data into k mutually exclusive subsets, each approximately equal size
 - At i -th iteration, use D_i as test set and others as training set
 - The mean of measures obtained in iterations used as the output of performance measure



Overfit

- One of the most common objective for cross validation is to avoid overfitting
- If a system is too complex, the performance of the system could be too good to be true



Q1. AI Lifecycle and Iteration (Ordering + Selection)

The lecture describes an AI system workflow with iterative refinement. Arrange the stages in the correct order:

- A. Collect Data
- B. Feature Extraction
- C. Machine Learning
- D. System Deployed
- E. Formulate Problem

Write the correct order as a sequence (e.g., $E \rightarrow A \rightarrow \dots$).

$E \rightarrow C \rightarrow A \rightarrow D \rightarrow B$

That is:

Formulate Problem $\rightarrow$ Collect Data $\rightarrow$ Feature Extraction $\rightarrow$ Machine Learning $\rightarrow$ System Deployed

Q2. Concerns vs Practices (Matching)

Match each problem to the most directly corresponding practice listed in the lecture

Problems:

1. Data privacy
2. Data collection bias (background/population mismatch)
3. False media
4. Power consumption of the system

Practices:

- A. Deepfake detection
- B. Anonymisation (really work?) / federated learning idea
- C. Know your population (particularly background information)
- D. Algorithm simplification

Write: 1-_____, 2-_____, 3-_____, 4-_____.

Q10. Security Threats to AI Systems (Classification)

The lecture lists actions that can mislead an AI system.

Classify each scenario as **Perturbation** or **Data poisoning**.

1. Adding small, visually imperceptible noise to an image at test time to flip the prediction: _____
2. Injecting a small number of maliciously crafted training samples into the dataset to degrade performance: _____
3. Adding barely audible noise to an audio clip at inference time to cause mis-transcription: _____
4. Planting mislabeled samples into the training set so a trigger pattern causes a targeted misclassification: _____

Q6. Domain Shift Responses (Matching)

Match each term to the best description according to the lecture context.

Terms:

- A. Domain generalisation
- B. Domain adaptation
- C. Transfer learning

Descriptions:

1. Learn from a source domain and adjust to a target domain with distribution shift.
2. Aim to perform well on unseen domains without direct target-domain tuning.
3. Reuse knowledge/representations from one task/domain to help learning in another.

Write: A-_____, B-_____, C-_____.

Q2. Intelligence Category (Matching)

Match each example to the most appropriate category.

Categories:

- A. Computation Intelligence (Feature $\rightarrow$ Label, unified)
- B. Perception Intelligence (Feature $\rightarrow$ Labels, non-unified)
- C. Creative Intelligence (beyond current generative models)

Examples:

1. Outlier detection
2. Emotion detection
3. Creative writing / story generation

Write: 1-_____, 2-_____, 3-_____.

Q3. Computer Vision Tasks vs Applications (Matching)

Match each application to the most essential computer vision task.

Tasks:

- A. Object detection/recognition
- B. Image segmentation
- C. Motion analysis
- D. Scene understanding (Q&A system)

Applications:

- 1. Autonomous vehicles
- 2. Medical imaging (tumour boundary delineation)
- 3. Surveillance systems (tracking movement in a scene)
- 4. Visual question answering over images

Write: 1-_____, 2-_____, 3-_____, 4-_____.

Q6. Audio Tasks and Methods (Matching)

Match each task to a method that is explicitly associated with it in the lecture content.

Methods:

- A. MFCCs
- B. HMMs
- C. NMF
- D. RNN/LSTM
- E. Transformer model (e.g., Whisper)
- F. Diffusion models

Tasks:

- 1. Speech feature extraction
- 2. Sequential audio modelling (classic approach)
- 3. Source separation
- 4. Temporal dependencies for speech-to-text
- 5. Multilingual ASR (modern approach)
- 6. High-fidelity music synthesis

Write: 1-_____, 2-_____, 3-_____, 4-_____, 5-_____, 6-_____.